

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

**Duration : 1 Hour
Total Marks:30**

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Shanmuganathan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Syde
Signature of the student:

(89)
HP

Problem 1:-

Nurse Scheduling with Backtracking Search

Variable, Domains and Constraints

We first formally define the system algorithm using standard AI triplet notation: variable (x), Domains (D) and Constraints (c).

* Variable (x): The entities requiring assignment

$$x = \{N_1, N_2, N_3\}$$

* Domains (D): The allowable discrete values for each variable

$$D(N_i) = \{\text{ICU, General, Emergency}\} \forall i \in \{1, 2, 3\}$$

For mathematical evaluation of strict inequality constraints, we project the ordered categorical domains onto an integer scale.

ICU $\rightarrow$ 1 General $\rightarrow$ 2 Emergency $\rightarrow$ 3

* Constraints (c):

* unary constraints (c_1): $N_1 \neq 3$ (Nurse 1 cannot be assigned to Emergency)

- * Binary Constraint (C_2): $N_2 < N_3$ (Nurse 2 must be stationed in a lower tier ward than Nurse 3)
- * Global Alldifferent Constraint (C_3): $N_1 \neq N_2$, $N_2 \neq N_3$, $N_1 \neq N_3$
(Exactly one nurse per ward)
- * Unary Constraint (C_4): $N_3 \neq 1$ (Nurse 3 cannot be assigned to ICU)
- * Completeness (C_5): All variable must be assigned a non null domain value

Step 1: Instantiation of N_1 :

- * Action: Assign $N_1 = 1$ (ICU)
- * Constraint Check: Evaluate C_1 : $1 \neq 3$. Satisfied
- * State: Partial Assignment $\{N_1 = 1\}$

Step 2: Instantiation of N_2

- * Attempt 2a: Assign $N_2 = 1$ (ICU)
- * Constraint Check: Evaluate C_3 : $N_1 = N_2 = 1$. Violated
- * Action: Prune Branch. Back-track.

3) Final Valid Schedule Profile

Variable (Nurse)	Ward	Integer Code
1	ICU	1
2	General	2
3	Emergency	3

Problem 2: Cleaning Robot Navigation via Depth limited Search (DLS)

1) Algorithm Operational Parameters:

* State Representation:

Each node is tracked using a 4 tuple Vector: $S = (x, y, d, c)$, where $(x, y) \in [1, 5] \times [1, 5]$ denotes discrete grid space coordinates, d is current depth level and c is accumulated battery utilization cost.

* Depth limit (L): $L = 8$ Any node generated

where $d > 8$ is structural failure: Expansion Stop

- * Attempt 2b: Assign $N_2 = 2$ (Correct)
- * Constraint: Evaluate $C_3: 1 \neq 2$. Satisfied
- * State: Partial assignment $\{N_1 = 1, N_2 = 2\}$

Step 3: Instantiation of N_3

- * Attempt 3a: Assign $N_3 = 1$ (ICU)
- * Constraint Check: Evaluate unary constraint $C_4 (N_3 \neq 1)$ Violated
- * Action: Prune Branch. Backtrack
- * Attempt 3b: Assign $N_3 = 2$ (Correct)
- * Constraint: Evaluate $C_3: N_2 = N_3 = 2$. Violated
- * Action: Prune branch. Backtrack
- * Attempt 3c: Assign $N_3 = 3$ (Emerging)
- * Constraint: Check
- * Evaluate: $[2 \text{ word}(N_2) \wedge \text{Word}(N_3) \rightarrow 2 < 3]$. Satisfied
- * Evaluate: $C_3: 1 \neq 2 \neq 3$. Satisfied
- * State: Complete assignment $\{N_1 = 1, N_2 = 2, N_3 = 3\}$

* Operator:

Up $(y+1)$ $\rightarrow$ Down $(y-1)$ $\rightarrow$ Left $(x-1)$ $\rightarrow$ Right $(x+1)$

* Graph Style Cycle Pursing Rules:

A potential Successor node (x', y') is blocked if its Spatial coordinates match any parent node residing in the path directly connecting the current leaf back to the root node.

Deep path penetration:

1) Level 0 to 3 The algorithm drives straight upward through the grid. At nodes 3 (2,5) exiting the up operator attempts to reference coordinate (2,6). This fails structural environment.

2) Cycle Elimination:-

At node 3, the second priority operator is Down, which reference (2,4). Since (2,4) is identical to node 2 this transition is pruned to prevent local loops.

3) Horizontal Transversal:

The engine shifts priority to third operator
Left Successful creating node 4 at coordinate (1,5)

4) Termination at Boundary Limit:

Expansion continue down this path. If the engine reaches a node where $d=8$ and the spatial coordinate does not equal the goal G , the branch is clipped. The search engine discard the node and steps backward to depth level 7 to try the next legal operator.

* Algorithm Conclusion

* Condition A: Valid path Exist if a target node matches G within the horizon, DLS isolates the sequence. The total battery cost will exactly equal the target's depth index value ($c=d \leq 8$)

* Condition B: No Solution Found: If the search space up to depth $d=8$ is completely exhausted without hitting G , the search prove no valid path exists within the operating Specification of the battery.